CMPT 455 Data Mining

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Introduction

Introduction

Contents of this Chapter

Overview of the course

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Big Data, Data Science, Data Mining

Relationship to other disciplines

Knowledge discovery process

Overview of the Course

Prerequisites

Basic Algorithms

- algorithms,
- data structures,
- complexity,
- efficiency.

Basic Statistics

- random variables, means, standard deviation,
- probability, probability distributions, Bayes theorem,
- sampling.

Overview of the Course

Learning Outcomes

- Understanding of the main KDD concepts
- Knowledge of the most important data mining tasks and algorithms
- Practical skills: implementing data mining algorithms and using state-of-the-art data mining tools
- Ability to select and implement data mining methods for a given application
- Foundation for research on new data mining methods

Overview of the Course

Outline

1. Introduction
2. Data preprocessing
3. Classification
4. Cluster analysis
5. Outlier detection
6. Association rules and frequent pattern mining
7. Impact of data mining
8. Research topics

References

Textbook

- Aggarwal, C.: „*Data Mining: The Textbook*“, Springer, 2015.
- http://rd.springer.com/book/10.1007/978-3-319-14142-8

Further recommended books

- Han J., Kamber M., Pei J.: „*Data Mining: Concepts and Techniques*“, Morgan Kaufmann Publishers, 3rd ed., 2011.
- Leskovec J., Rajaraman A., Ullman J.D.: „*Mining of Massive Datasets*“, Cambridge University Press, 2nd ed., 2014.

Research articles

on demand

References

Other resources

- KDNuggets: a very comprehensive resource of KDD software, companies, publications and more. http://www.kdnuggets.com/
- ACM SIGKDD: ACM's special interest group on Knowledge Discovery in Databases. http://www.acm.org/sigkdd/

Open source data mining tools

- Python (Scikit-learn): https://scikit-learn.org/stable/
- R: https://www.r-project.org/

Tentative Grading Scheme

Course project

Data mining on a real-life dataset of your choice using state-of-the-art tools

Group project (2 or 3 students)

Project proposal including discussion of related work	10%
Data preprocessing and exploratory data analysis	10%
Project report and code (Github site)	20%

Exams

Test knowledge of key concepts and algorithms

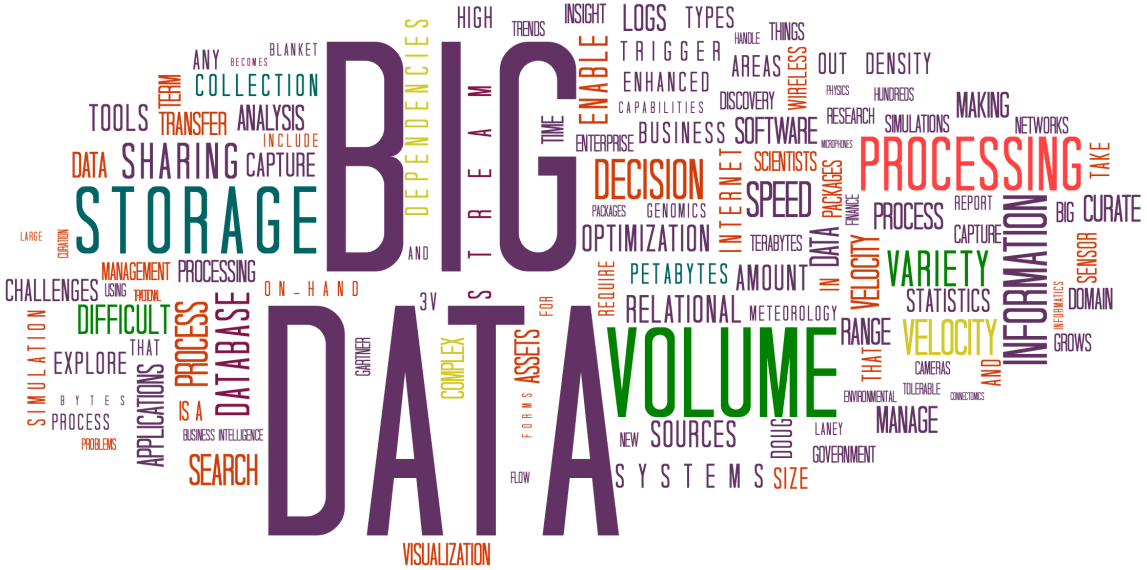
Midterm (1.5 hours)	20%
Final (3 hours)	40%

Students must attain an overall passing grade on the weighted average of exams in the course in order to obtain a clear pass (C- or better).

Big Data, Data Science, Data Mining

Big Data

- Massive volume of both structured and unstructured data that is too large to be processed using traditional database and software techniques.



- Technology that an organization requires to store and analyze the large amounts of data.

Big Data, Data Science, Data Mining

Big Data Applications



Facebook

- Captures approximately 4 PB (10^{15}B) user data daily.

On-platform activity — posts, likes, shares, comments, searches, watch time, who you interact with

Device and technical data — IP address, browser type, device identifiers, operating system

Location data — GPS location, inferred location from IP

Messages — metadata from Messenger (and in some cases content, depending on settings)

Ad interaction data — which ads you see, click, and how long you linger on them

- Recommends friend and items, makes targeted ads.

Big Data, Data Science, Data Mining

Big Data Applications



Amazon

- Collects 5-20 EB (10^{18}B) of user data daily.

Amazon.com (shopping)

Search queries, browse history, purchase history, reviews, ad clicks

Alexa / Echo

Voice recordings, smart home usage, routines, location

Prime Video

Watch history, pause/rewind behavior, ratings

Kindle / Audible

Reading behavior, highlights, listening speed

- Recommends items, makes targeted ads.

Big Data, Data Science, Data Mining

Big Data Applications

Large Hadron Collider



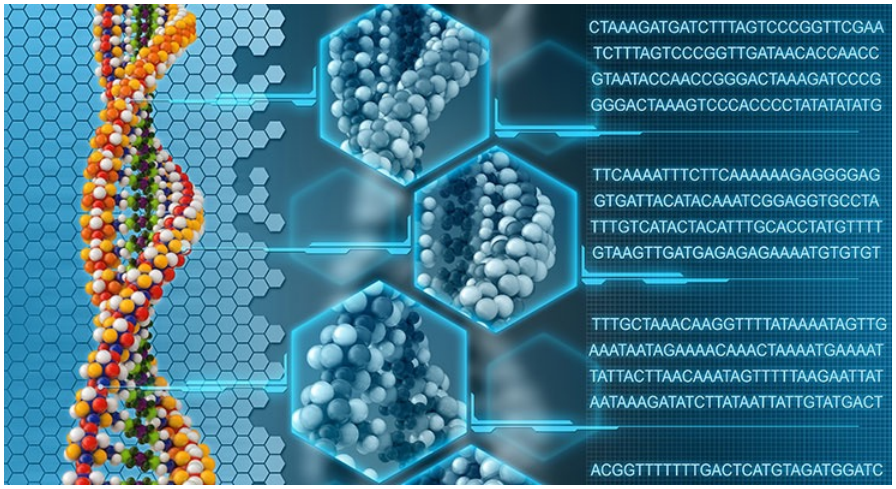
- For particle physics experiments, the largest experimental facility in the world.
- About 1PB collision data per second, 84EB per day.
- Detects interesting (anomalous) collisions in real time and keeps 1PB per day.
- Detect and characterize new particles.

Big Data, Data Science, Data Mining

Big Data Applications

Genomics

- Sequencing one human genome produces ~1TB of data.
- Precision Medicine Initiative: 1 million volunteers provide EHRs, healthcare claims, biological samples (e.g. for DNA collection).



- Goal: understand relationship between genotype and phenotype for more precise, personalized diagnostics and treatment.

Big Data, Data Science, Data Mining

Data Science

What is it?

- Understanding the past data and predicting the future to gain actionable insight for an organization.



Is it just a new, fancy term for Statistics?

Requirements

- Not only technical skills,
- but also domain knowledge and communication skills.

High demand

- starting salary at the entry level ~ $80K

Big Data, Data Science, Data Mining

Important competencies

Table 3. Important competencies for data analytics.

Consensus	Cognitive	Intrapersonal	Interpersonal	Responses (N)	Important/ Essential (N)	Consensus
Iteration #1						
Analysis	•			64	62	**97%**
Continuous learning		•		65	61	**94%**
Problem solving	•			65	60	**92%**
Reasoning/ argumentation	•			64	58	**91%**
Critical thinking	•			64	57	**89%**
Trust			•	64	57	**89%**
Intellectual interest		•		65	56	**86%**
Responsibility		•		65	56	**86%**
Collaboration cooperation			•	65	56	**86%**
Personal responsibility		•		62	53	**85%**
Professionalism		•		62	53	**85%**
Share information			•	60	51	**85%**
Communication			•	64	54	**84%**
Commitment to trustworthiness			•	64	54	**84%**
Interpretation	•			62	52	**84%**
Teamwork			•	65	54	**83%**
Communication non-technical			•	65	54	**83%**

Big Data, Data Science, Data Mining

Non-important competencies

Table 4. Non-important competencies for data analytics.

Consensus	Cognitive	Intrapersonal	Interpersonal	Responses (N)	Important/ Essential (N)	Consensus
Iteration #1						
Risk-taking		•		61	14	**23%**
Physical health		•		55	11	**20%**
Career orientation		•		63	12	**19%**
Artistic		•		58	7	**12%**

Definition KDD

Knowledge discovery in databases (*KDD*) is the process of (semi-)automatic extraction of knowledge from databases which is

- *valid,*
- *previously unknown,* and
- *potentially useful*.

Remarks

- *(semi)-automatic*: different from manual analysis.
  Often, some user interaction is necessary.
- *valid*: in the statistical sense.
- *previsouly unknown*: not explicit, no „common sense knowledge“.
- *potentially useful*: for a given application.

Relationship to Other Disciplines

Contributions from Database Systems

- scalability for large datasets
- integration of data from different sources (data warehouses)
- novel datatypes (e.g. text and web data)

Contributions from Statistics

- probabilistic knowledge
- model-based inferences
- evaluation of knowledge

Contributions from Machine Learning

- different paradigms of learning
- supervised learning
- hypothesis spaces and search strategies

Relationship to Other Disciplines

Database Systems

+ discovery of implicit (not explicit) patterns
+ learning capabilities

Statistics

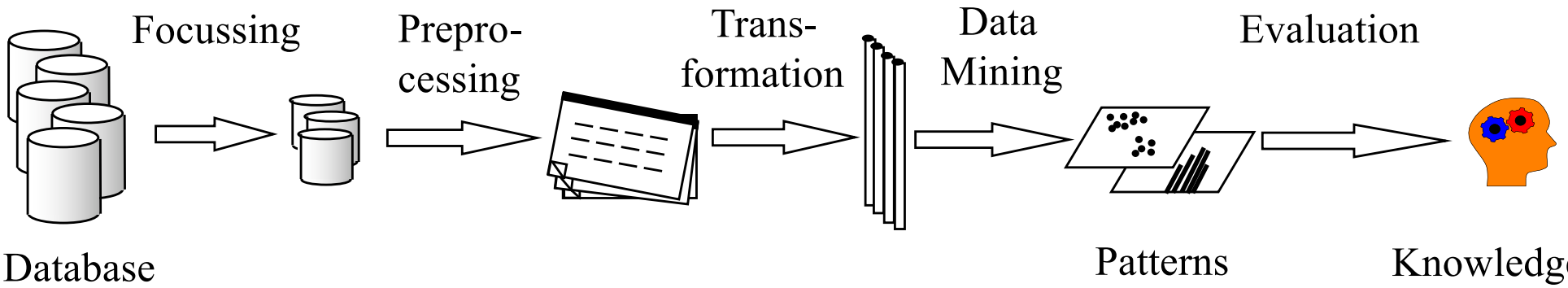
+ analysis of existing databases (not designed for task)
+ automatic generation of plausible hypotheses
+ efficient algorithms

Machine Learning

+ dealing with imperfect data
+ very large datasets
+ understandability of knowledge

KDD Process

KDD Process Model





iterative and
interactive process

Focussing

Understanding the application

Ex.: make new telecommunication rates

Definition of the KDD goal

Ex.: customer segmentation

Data acquisition

Ex.: from operational billing DB

Data management

file system or DBS?

Selection of relevant data

Ex.: 100‘000 important customers with all calls in 2021

Preprocessing

Integration of data from different sources

- Simple conversion of attribute names (e.g. CNo → CustomerNumber)
- Use of domain knowledge for duplicate detection (e.g. spatial match based on ZIP codes)

Consistency check

- Test of application specific consistency constraints
- Resolution of inconsistencies

Completion

- Substitution of unknown attribute values by defaults
- Distribution of attribute values shall not be changed

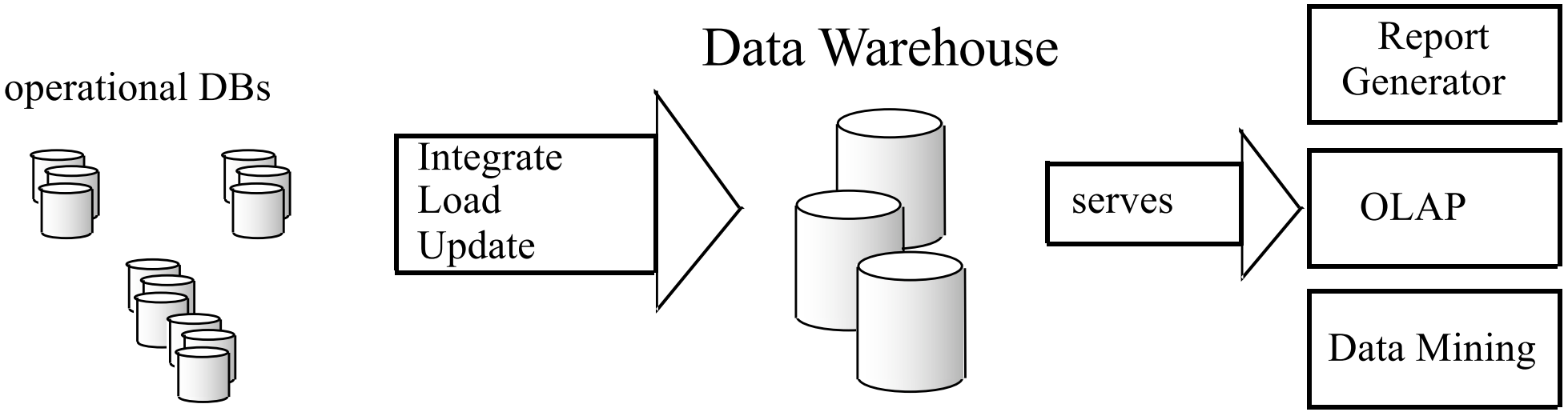


preprocessing is often the most expensive KDD step

Preprocessing

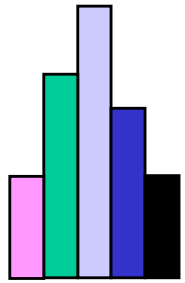
Data Warehouse

- persistent
- integrated collection of data
- from different sources
- for the purpose of analysis or decision support



Transformation

Discretization of numerical attributes



- Independent from the data mining task

  Ex.: partitioning of the attribute domain in equal-length intervals

- Specific for the data mining task

  Ex.: partitioning in intervals such that the information gain w.r.t. class membership is maximized

Generation of derived attributes

- Aggregation over sets of data records

  Ex.: from single call records to

  „Total minutes daytime / evening, weekday / weekend“

- Combination of several attributes

  Ex.: minutes change = total minutes 2024 – total minutes 2023

Transformation

Selection of attributes (features)

- *manual*
  if domain knowledge available on the attribute semantics and on the data mining task
- *automatic*
  bottom-up (starting from the empty set, add one attribute at a time)
  or top-down
  (starting from the set of all attributes, remove one attribute at a time)
  e.g. optimizing the discrimination between the different classes

⇨ too many attributes can lead to inefficient and ineffective data mining

⇨ some transformations can be realized by OLAP-systems

Data Mining

Definition

Data Mining is the application of efficient algorithms that determine the patterns contained in a database.

Data mining tasks



clustering *classification*

A and B → C

association rules

other tasks: regression, outlier detection . . .

Data Mining

Applications

- Clustering

  customer segmentation, structuring sets of web documents,

  determining protein families and superfamilies

- Classification

  automatic credit check, automatic interpretation of astronomical images,

  prediction of drug response

- Association rules

  redesign of supermarket layout, improving cross-selling,

  recommender systems

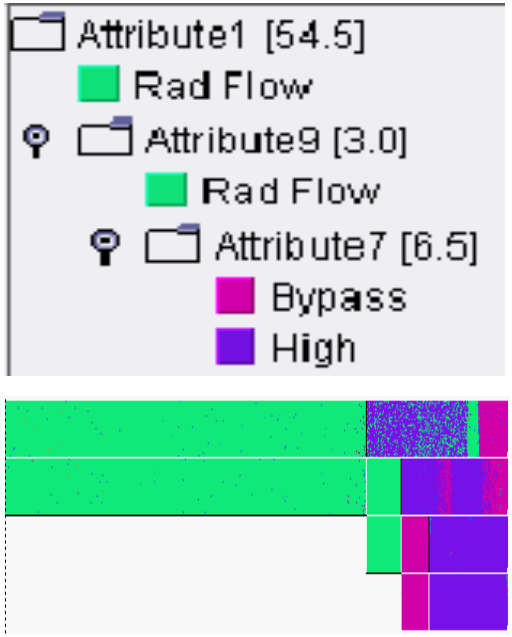
Evaluation

Procedure

- Presentation of discovered patterns supported by appropriate visualizations
- Evaluation of the patterns by the user
- If evaluation not satisfactory: repeat data mining with
  - Different parameters
  - Different methods
  - Different data
- If evaluation o.k.:

  Integration of discovered knowledge in the enterprise knowledge base

  Use of the new knowledge for future KDD-processes



Evaluation

Evaluation of discovered patterns

Interestingness

- Pattern already known?
- Pattern surprising?
- Pattern relevant for the application?

Predictive power

- How accurate is the pattern? (*confidence*)
- For how many cases does the pattern apply? (*support*)
- How well does the pattern generalize to unseen cases?